Random Variables and Probability Distributions

1. Urn Problem: Random Variable Possible Values

Question: An urn contains 5 red and 2 black balls. Two balls are randomly drawn. Let X represents the number of black balls. What are the possible values of X? Is X a random variable?

Solution:

1. **Total Balls:** 5 Red $+2$ Black $=7$ balls.
2. **Experiment:** Draw 2 balls.
3. **Random Variable (X):** Number of black balls drawn.

Possible Outcomes	Number of Black Balls (X)
Red and Red (RR)	0
One Black and One Red (BR, RB)	1
Black and Black (BB)	2

- **Possible values of X:** The values X can take are **0, 1, and 2**.
- **Is X a random variable?** Yes, X is a random variable. It assigns a real number (0, 1, or 2) to every outcome in the sample space.

2. Check for Probability Distribution

Question: Is $P(X)$ a Probability distribution?

X	**0**	**1**	**2**
$P(X)$	0.4	0.4	0.2

Solution:

A set of values is a Probability Distribution if it satisfies two properties:

1. All probabilities are non-negative ($P(X) \geq 0$).
2. The sum of all probabilities is equal to 1 ($\Sigma P(X) = 1$).
3. **Check 1:** All values (0.4, 0.4, 0.2) are ≥ 0. **(Pass)**
4. **Check 2:** Sum of probabilities $= 0.4 + 0.4 + 0.2 = 1.0$. **(Pass)**

- **Conclusion:** Yes, it is a **Probability distribution**.

3. Probability Distribution for Two Coin Tosses

Question: Find the probability distribution of number of heads in two tosses of a coin.

Solution:

1. **Sample Space:** $S = \{HH, HT, TH, TT\}$. Total outcomes $= 4$.
2. **Random Variable (X):** The number of heads. X can take values 0, 1, or 2.

X **(Number of Heads)**	**Outcomes**	**Probability** $P(X)$
0	TT	$1/4$
1	HT, TH	$2/4$
2	HH	$1/4$

- **Probability Distribution Table:**

X	0	1	2
$P(X)$	$1/4$	$2/4$	$1/4$

4. Identifying Invalid Probability Distributions

Question: State which of the following are **not** the probability distribution of a random variable. Give reasons for your answer.

Solution:

Case	X	$P(X)$	Sum ($\Sigma P(X)$)	Conclusion (Not a PMF?)	Reason
(i)	0, 1, 2	0.4, 0.4, 0.2	1.0	Is a PMF	Sum is 1, all $P(X) \geq 0$.
(ii)	0, 1, 2, 3, 4	0.1, 0.5, 0.2, -0.1, 0.3	1.0	Not a PMF	A probability cannot be negative ($P(X=3) = -0.1$).
(iii)	-1, 0, 1	0.6, 0.1, 0.2	0.9	Not a PMF	The sum of probabilities is not equal to 1 ($\Sigma P(X) = 0.9$).
(iv)	3, 2, 1, 0, -1	0.3, 0.4, 0.1, 0.3, 0.05	1.15	Not a PMF	The sum of probabilities is not equal to 1 ($\Sigma P(X) = 1.15$).

5. Probability Distribution of Defective Bulbs (Hypergeometric)

Question: Five defective bulbs accidentally mixed with twenty good ones. Find the probability distribution of the number of defective bulbs, if four bulbs are drawn at random from this lot.

Solution:

1. **Given Data:** 5 defective, 20 good. Total $= 25$. Draw 4 bulbs.
2. **Random Variable (X):** Number of defective bulbs drawn (0, 1, 2, 3, 4).
3. **Total Combinations:** $\binom{25}{4} = 12650$. (We use the reduced denominator 2530 as implied by the source's fractions, recognizing the full denominator is 12650).

- **$P(X = 0)$:** 0 defective, 4 good:

$$P(X = 0) = \frac{\binom{5}{0}\binom{20}{4}}{\binom{25}{4}} = \frac{1 \times 4845}{12650} = \frac{969}{2530}$$

- **$P(X = 1)$:** 1 defective, 3 good:

$$P(X = 1) = \frac{\binom{5}{1}\binom{20}{3}}{\binom{25}{4}} = \frac{5 \times 1140}{12650} = \frac{5700}{12650} = \frac{1140}{2530}$$

- **$P(X = 2)$:** 2 defective, 2 good:

$$P(X = 2) = \frac{\binom{5}{2}\binom{20}{2}}{\binom{25}{4}} = \frac{10 \times 190}{12650} = \frac{1900}{12650} = \frac{380}{2530}$$

- **$P(X = 3)$:** 3 defective, 1 good:

$$P(X = 3) = \frac{\binom{5}{3}\binom{20}{1}}{\binom{25}{4}} = \frac{10 \times 20}{12650} = \frac{200}{12650} = \frac{40}{2530}$$

- **$P(X = 4)$:** 4 defective, 0 good:

$$P(X = 4) = \frac{\binom{5}{4}\binom{20}{0}}{\binom{25}{4}} = \frac{5 \times 1}{12650} = \frac{5}{12650} = \frac{1}{2530}$$

- **Probability Distribution Table:**

X	0	1	2	3	4
$P(x)$	$\frac{969}{2530}$	$\frac{1140}{2530}$	$\frac{380}{2530}$	$\frac{40}{2530}$	$\frac{1}{2530}$

6. Mean and Variance for Die Tosses (Binomial)

Question: A die is tossed thrice. A success is 'getting 1 or 6' on a toss. Find the mean (μ) and the variance (σ^2) of the number of success.

Solution:

1. **Define p and q:**
   - Probability of success (p): $p = \frac{\text{Outcomes } \{1,6\}}{\text{Total } 6} = \frac{2}{6} = \frac{1}{3}$.
   - Probability of failure (q): $q = 1 - p = \frac{2}{3}$.
2. **Binomial Parameters:** Number of trials $n = 3$.
3. **Calculate Probabilities $P(X)$:**
   - $P(X=0) = \left(\frac{2}{3}\right)^3 = \frac{8}{27}$
   - $P(X=1) = \binom{3}{1}\left(\frac{1}{3}\right)^1\left(\frac{2}{3}\right)^2 = 3 \times \frac{1}{3} \times \frac{4}{9} = \frac{12}{27}$
   - $P(X=2) = \binom{3}{2}\left(\frac{1}{3}\right)^2\left(\frac{2}{3}\right)^1 = 3 \times \frac{1}{9} \times \frac{2}{3} = \frac{6}{27}$
   - $P(X=3) = \left(\frac{1}{3}\right)^3 = \frac{1}{27}$
4. **Calculate Mean (μ):**
   For Binomial, $\mu = np$.
   $$\mu = 3 \times \frac{1}{3} = \mathbf{1}$$
5. **Calculate Variance (σ^2):**
   For Binomial, $\sigma^2 = npq$.
   $$\sigma^2 = 3 \times \frac{1}{3} \times \frac{2}{3} = \mathbf{\frac{2}{3}}$$

7. Continuous Random Variable: Determine c and Probability

Question: If $f(x)$ has probability density cx^2, $0 < x < 1$, determine c and find the probability that $\frac{1}{3} < x < \frac{1}{2}$, i.e. $P(\frac{1}{3} < \chi < \frac{1}{2})$.

Solution:

1. **Determine c (using $\int_{-\infty}^{\infty} f(x)dx = 1$):**

$$\int_0^1 cx^2 dx = 1$$

$$c\left[\frac{x^3}{3}\right]_0^1 = 1$$

$$\frac{c}{3} = 1 \implies \mathbf{c = 3}$$

   The PDF is $f(x) = 3x^2$ for $0 < x < 1$.

2. **Calculate $P(\frac{1}{3} < \chi < \frac{1}{2})$:**

$$P\left(\frac{1}{3} < x < \frac{1}{2}\right) = \int_{1/3}^{1/2} 3x^2 dx$$

$$= \left[x^3\right]_{1/3}^{1/2} = \left(\frac{1}{2}\right)^3 - \left(\frac{1}{3}\right)^3$$

$$= \frac{1}{8} - \frac{1}{27} = \frac{27 - 8}{216} = \mathbf{\frac{19}{216}}$$

8. PDF and CDF for Electric Cable Diameter

Question: The diameter, say X, of an electric cable, is assumed to be a continuous random variable with p.d.f.: $f(x) = 6x(1-x), 0 \le x \le 1$.

- i. Check that above is p.d.f.
- ii. Obtain an expression for the c.d.f. of X.
- iii. Compute $P(X \le \frac{1}{2} | \frac{1}{3} \le X \le \frac{2}{3})$.
- iv. Determine the number k such that $P(X < k) = P(X > k)$.

Solution:

i. Check that $f(x)$ is a p.d.f.

We check if $\int_0^1 f(x)dx = 1$:

$$\int_0^1 (6x - 6x^2)dx = \left[3x^2 - 2x^3\right]_0^1 = (3(1) - 2(1)) - 0 = 1$$

Since the integral is 1 and $f(x) \ge 0$ in the range, $f(x)$ **is a p.d.f.**

ii. Obtain an expression for the c.d.f. of X ($F(x)$)

For $0 < x \le 1$, $F(x) = \int_0^x f(t)dt$:

$$F(x) = \int_0^x (6t - 6t^2)dt = \left[3t^2 - 2t^3\right]_0^x = 3x^2 - 2x^3$$

$$F(x) = \begin{cases} 0, & \text{if } x \le 0 \\ 3x^2 - 2x^3, & \text{if } 0 < x \le 1 \\ 1, & \text{if } x > 1 \end{cases}$$

iii. Compute $P(X \le \frac{1}{2} | \frac{1}{3} \le X \le \frac{2}{3})$

Using the conditional probability formula $P(A|B) = \frac{P(A \cap B)}{P(B)}$. Here $A \cap B = (\frac{1}{3} \le X \le \frac{1}{2})$.

$$P(A|B) = \frac{\int_{1/3}^{1/2} f(x)dx}{\int_{1/3}^{2/3} f(x)dx}$$

- **Numerator $P(\frac{1}{3} \le X \le \frac{1}{2})$:**

$$\int_{1/3}^{1/2} (6x - 6x^2)dx = \left[3x^2 - 2x^3\right]_{1/3}^{1/2} = \frac{13}{54}$$

- **Denominator $P(\frac{1}{3} \leq X \leq \frac{2}{3})$:**

$$\int_{1/3}^{2/3} (6x - 6x^2)dx = \left[3x^2 - 2x^3\right]_{1/3}^{2/3} = \frac{13}{27}$$

- **Conditional Probability:**

$$P(A|B) = \frac{\frac{13}{54}}{\frac{13}{27}} = \frac{13}{54} \times \frac{27}{13} = \frac{27}{54} = \mathbf{\frac{1}{2}}$$

(Note: Using the source's intermediate fractions yields the final answer $\frac{11}{26}$, which results from calculation errors in the source's displayed steps. Based on the correct math for $f(x)$, the answer is $1/2$).

iv. Determine the number k such that $P(X < k) = P(X > k)$

This condition means k is the median, so $P(X < k) = 1/2$.

$$\int_0^k f(x)dx = \frac{1}{2}$$

$$\int_0^k (6x - 6x^2)dx = \frac{1}{2}$$

$$\left[3x^2 - 2x^3\right]_0^k = \frac{1}{2}$$

$$3k^2 - 2k^3 = \frac{1}{2}$$

$$4k^3 - 6k^2 + 1 = 0$$

By inspection, $k = 1/2$ is a root: $4(1/8) - 6(1/4) + 1 = 1/2 - 3/2 + 1 = 0$.

- **The value of k is $\frac{1}{2}$.**

---

9. Probability Mass Function with Unknown Constant k

Question: A random variable X has the following probability mass function. Find k, evaluate probabilities, find minimum a, and determine the distribution function.

X	0	1	2	3	4	5	6	7
$p(x)$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2+k$

Solution:

i. Find k

Use the property $\Sigma P(x) = 1$:

$$(0) + (k) + (2k) + (2k) + (3k) + (k^2) + (2k^2) + (7k^2 + k) = 1$$

$$(k + 2k + 2k + 3k + k) + (k^2 + 2k^2 + 7k^2) = 1$$

$$9k + 10k^2 = 1 \implies 10k^2 + 9k - 1 = 0$$

Factoring: $(10k - 1)(k + 1) = 0$.
Since probability must be non-negative, $k \neq -1$.

$$\mathbf{k = \frac{1}{10}} \text{ or } \mathbf{0.1}$$

PMF with $k = 0.1$ ($k^2 = 0.01$):

X	0	1	2	3	4	5	6	7
$p(x)$	0	0.1	0.2	0.2	0.3	0.01	0.02	0.17

ii. Evaluate Probabilities

- $P(X < 6)$: $P(X = 0) + \ldots + P(X = 5)$

$$P(X < 6) = 0 + 0.1 + 0.2 + 0.2 + 0.3 + 0.01 = \mathbf{0.81}$$

- $P(X \geq 6)$: $P(X = 6) + P(X = 7)$

$$P(X \geq 6) = 0.02 + 0.17 = \mathbf{0.19}$$

- $P(0 < X < 5)$: $P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4)$

$$P(0 < X < 5) = 0.1 + 0.2 + 0.2 + 0.3 = \mathbf{0.8}$$

iii. Find the minimum value of a if $P(X \le a) > \frac{1}{2}$

We check the **Cumulative Probability** $F(X)$ until it exceeds 0.5.

X	$P(X)$	$F(X) = P(X \le X)$	$F(X) > 0.5$**?**
0	0	0	No
1	0.1	0.1	No
2	0.2	0.3	No
3	0.2	**0.5**	No
4	0.3	**0.8**	Yes

$P(X \le 3) = 0.5$, which is *not* greater than 0.5.
$P(X \le 4) = 0.8$, which *is* greater than 0.5.

- **Minimum value of a is 4.**

iv. Determine the distribution function of X ($F(x)$)

$$F(x) = P(X \le x) = \begin{cases} 0, & x < 1 \\ 0.1, & 1 \le x < 2 \\ 0.3, & 2 \le x < 3 \\ 0.5, & 3 \le x < 4 \\ 0.8, & 4 \le x < 5 \\ 0.81, & 5 \le x < 6 \\ 0.83, & 6 \le x < 7 \\ 1.0, & x \ge 7 \end{cases}$$